6. Fill in the following table so that the binary operation $*$ on $A=\{a,b,c\}$ is commutative.

$*$	a	b	c
a	b		
b	c	b	a
c	a		c

7. Consider the binary operation $*$ defined on the set $A=\{a,b,c,d\}$ by the following table:

$*$	a	b	c	d
a	a	c	b	d
b	d	a	b	c
c	c	d	a	a
d	d	b	a	c

Is it commutative and associative?

8. Let $A=\begin{pmatrix}1&0&1&0\\0&1&0&1\\1&0&0&1\end{pmatrix}$, $B=\begin{pmatrix}0&1&0&1\\1&0&1&0\\1&0&0&1\end{pmatrix}$, $C=\begin{pmatrix}1&1&0&1\\0&1&1&0\\1&1&1&1\end{pmatrix}$ be any three boolean matrices of the same type. Find (i) $A\vee B$ (ii) $A\wedge B$ (iii) $(A\vee B)\wedge C$ (iv) $(A\wedge B)\vee C$.

9. (i) Let $M=\left\{\begin{pmatrix}x&x\\x&x\end{pmatrix}: x\in R-\{0\}\right\}$ and let $*$ be the matrix multiplication. Determine whether M is closed under $*$. If so, examine the commutative and associative properties satisfied by $*$ on M.

(ii) Let $M=\left\{\begin{pmatrix}x&x\\x&x\end{pmatrix}: x\in R-\{0\}\right\}$ and let $*$ be the matrix multiplication. Determine whether M is closed under $*$. If so, examine the existence of identity, existence of inverse properties for the operation $*$ on M.

10. (i) Let A be $\mathbb{Q}\setminus\{1\}$. Define $*$ on A by $x*y=x+y-xy$. Is $*$ binary on A? If so, examine the commutative and associative properties satisfied by $*$ on A.

(ii) Let A be $\mathbb{Q}\setminus\{1\}$. Define $*$ on A by $x*y=x+y-xy$. Is $*$ binary on A? If so, examine the existence of identity, existence of inverse properties for the operation $*$ on A.

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